Hackathon Projects - Component Allocation Breakdown

Overview

This document details the allocation and bill of materials (BOM) for the student teams participating in the robotics hackathon, divided equally across three project challenges (5 teams per challenge). All hardware items are sourced from the Botland order.

Project 1: Small Robot That Measures Distance to Obstacles (5 Teams max)

Objective & Challenge Overview

Design and construct an autonomous mobile two-wheel drive (2WD) robotic rover capable of navigating dynamic environments by continuously measuring distances to surrounding obstacles and executing avoidance maneuvers to prevent collisions.

Component Allocation Table

Component	SKU / Model	Qty per Team
Arduino Uno R4 WiFi	ARD-23292	1 pc
2WD Robot Chassis Kit	ROB-07283 / FTC-09213	1 kit
Stepper Motor 28BYJ-48 5V + ULN2003 Driver	GRL-12807	2 sets
Ultrasonic Distance Sensor HC-SR04	MOD-01420	1 pc

Component	SKU / Model	Qty per Team
Micro Servo SG-90 (180°)	DNG-13128	1 pc
Cell Holder for 2x 18650 (Serial) OR power from laptop OR AA batteries	AKC-05240	1 pc
Breadboard (400 holes / 170 holes)	JUS-19942 / AKS-10288	1-2 pcs
justPi Dupont Jumper Cables (M-M, M-F, F-F)	Assorted lengths (10 cm / 20 cm)	1 pack

Project 2: Dual-Axis Solar Tracker (5 Teams max)

Objective & Challenge Overview

Build an automated dual-axis electro-mechanical tracking station that continuously aligns a solar panel toward the brightest source of light to optimize energy harvesting efficiency.

Component Allocation Table

Component	SKU / Model	Qty per Team
Arduino Uno R4 WiFi	ARD-23292	1 pc
Micro Pan/Tilt Servo Bracket	SPF-01948	1 set
Micro Servo SG-90 (180°)	DNG-13128	2 pcs
Photoresistors 10-20kΩ (GL5528)	SES-04598	4 pcs

Component	SKU / Model	Qty per Team
Solar Cell 0.7W / 5.5V (95x95mm)	AKU-03695	1 pc
DFRobot Gravity v2.1.0 I2C Digital Wattmeter	DFR-13753	1 pc
Rotary Potentiometer 10kΩ Linear	PAS-02168	1 pc
Cell Holder for 2x 18650 (Serial) OR power from laptop OR AA batteries	AKC-05240	1 pc
Breadboard justPi (830 holes)	JUS-19943	1 pc
10kΩ Fixed Resistors	From E32 / Iduino kits	4 pcs
justPi Dupont Jumper Cables (M-M, M-F, F-F)	Assorted lengths	1 pack

Project 3: Smart Trash Can (5 Teams)

Objective & Challenge Overview

Create an automated, contactless waste container prototype that senses user proximity to automatically open its lid for hygienic operation and monitors internal container fill levels.

Component Allocation Table

Component	SKU / Model	Qty per Team
Arduino Uno R4 WiFi	ARD-23292	1 pc
Ultrasonic Distance Sensor HC-SR04	MOD-01420	2 pcs
Servo TowerPro MG-996R (Robot 360 / High-Torque)	MOT-04798	1-2 pcs
Micro Servo SG-90 (180°)	DNG-13128	Alternate / Spare
Rotary Potentiometer 10kΩ Linear	PAS-02168	1 pc
Cell Holder for 2x 18650 (Serial) OR power from laptop OR AA batteries	AKC-05240	1 pc
Breadboard justPi (400 holes)	JUS-19942	1 pc
justPi Dupont Jumper Cables (M-M, M-F, F-F)	Assorted lengths	1 pack

Shared Prototyping Pool, Central Power Station & Spares Reserve

Prototyping Assortment Kits (Central Parts Organizers)

- **Set of electronic components - E32 (254 elements)** (SKU: GRL-12110): Comprehensive organizer with assorted resistors, LEDs, tactile switches, and capacitors.

- **Electronic components set for Arduino - Iduino KTS042** (SKU: OST-14319): Assortment of basic sensors, passive components, and jumper wires.
- **Electronic components set for Arduino - Iduino KTS021** (SKU: OST-14317): Includes 830-hole breadboard, displays, active sensors, and hardware.
- **Electronic components set AM402 + breadboard 400 (31 elements)** (SKU: GRL-13579): Portable prototyping set and breadboard.

Central Power & Recharging Station

- **10x Power cell 18650 Li-Ion XTAR - 2600mAh** (SKU: ZAS-09067): Shared rechargeable Li-ion battery pool for mobile and servo applications.
- **1x Battery LCD charger XTAR VC2 18650** (SKU: AKC-08542): Central dual-bay fast charger located at the organizer bench.

Jumper Wire Stockpile

- **118 bundles of 40-pin Dupont ribbons** (~4,720 cables total) across Female-Male, Male-Male, and Female-Female configurations in 10 cm and 20 cm lengths.

Buffer Spares Reserve

- **5x** Arduino Uno R4 WiFi boards (20 ordered vs. 15 team baseline)
- **5x** 2x 18650 Battery holders (20 ordered vs. 15 team baseline)
- **2x** Mini Solar cells 0.7W (7 ordered vs. 5 needed)
- **2x** DFRobot Gravity I2C Wattmeters (7 ordered vs. 5 needed)
- **2x** Stepper motors 28BYJ-48 + ULN2003 drivers (12 ordered vs. 10 needed)
- **3x** 2WD Robot chassis kits (8 ordered across Feetech, Rectangle, and Round models vs. 5 needed)
- **5x** Micro Servos SG-90 (20 ordered vs. 15 primary allocated)
- **30x** GL5528 Photoresistors (50 ordered vs. 20 needed)